RflySim components for R4R

Reuse PX4/CopterSim for stock motion and control integration.

Keep native RGB as a visual baseline. Two waypoint flights passed and B2 recorded RGB concurrently.

External motion replay is proposed. Native calibration and real-scene fidelity remain unverified.



Actual assets and versions

Software: Free v5.01-20260831. CopterSim 5.0.1, RflySim3D 5.01 / UE4.27.2, PX4 1.12.3dev.

Scene and vehicle: stock Grasslands, CopterSim model 0, Iris airframe 10016. Full physical model inventory is missing.

No real reconstruction, dataset or learned weights. “Checkpoint” ZIPs back up evidence. They are not learned models.

Native control and observation path



PX4 estimates, ACKs, modes and arming return to the supervisor.

Body truth, native camera metadata and image timestamps export separately.

Control UDP 20100/20101, SITL TCP 4560, truth UDP 30101, camera UDP 20006, RGB shared memory.

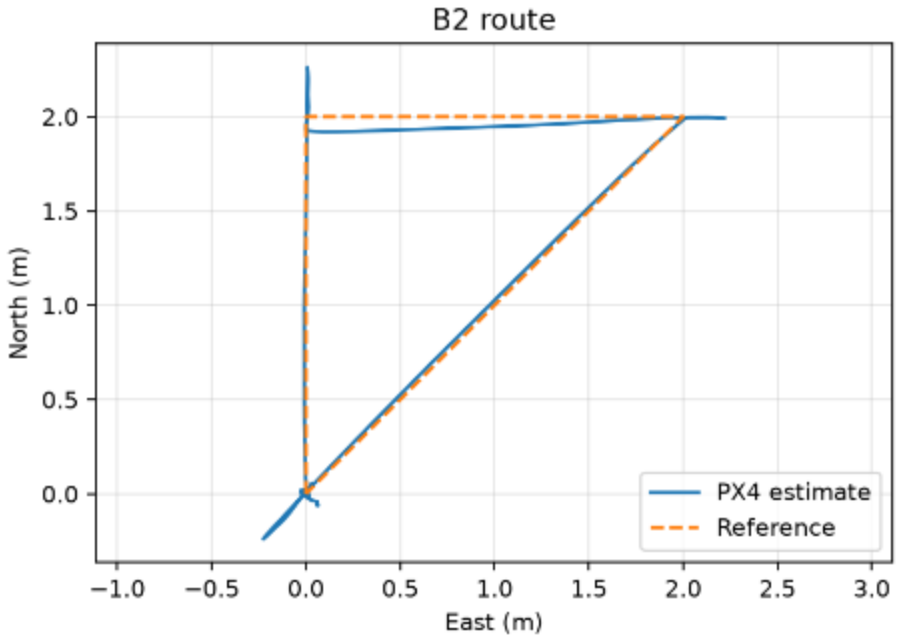
Algorithms and responsibilities

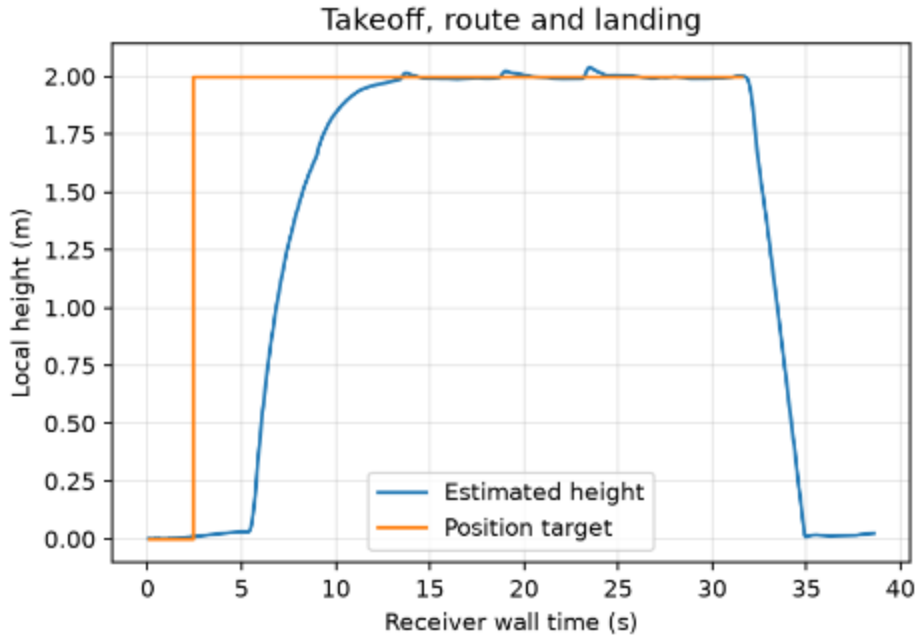
Our script advances a target only after fresh estimated position stays inside the tolerance for 2 seconds.

PX4 estimates motion and uses position, velocity, attitude and rate feedback to generate actuators. CopterSim integrates the stock dynamics.

UE renders scene pixels. A separate process copies frames and writes files. No image-based controller ran.

Both flights completed the route



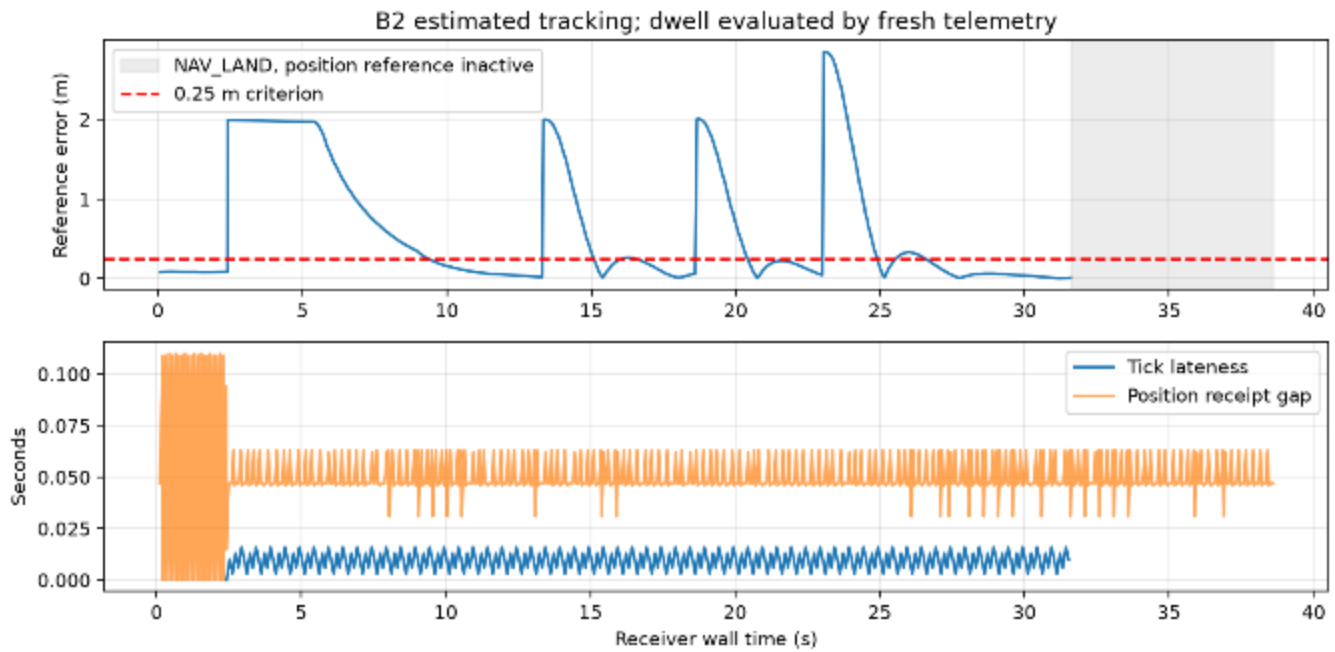


B1 without capture: 38.750 s. B2 with RGB: 39.219 s. Both minimum integration and full-route outcomes passed.

NED metres: (0,0,-2), (2,0,-2), (2,2,-2), (0,0,-2), land. Each leg required error ≤0.25 m for 2 s.

Fresh PX4 estimates provide the measurement basis. Landing and disarming are retained in the raw flight records.

Control timing with concurrent RGB



Mean target intervals: B1 100.022 ms, B2 100.034 ms. Maximum intervals: 109 ms and 110 ms.

Maximum tick lateness: 12 ms and 16 ms. Neither run exceeded 100 ms. Position gaps peaked at 125 ms and 110 ms.

These two sequential flights do not isolate capture overhead. Reset state, leg durations and scheduler timing also differ.

Playable native RGB recordings

A1: 260 frames, 26.0 s playback. Capture was short at 25.797 s and rotations were invalid.

A2: 261 frames, 26.1 s playback. Rotations were invalid. B2: 365 frames, 36.5 s playback during its route.

Rate test: 300 post-warmup frames, 10.013 Hz. Videos use fixed 10 FPS playback including warmup. Acquisition time is unknown.



A1 video A2 video B2 video

Worked record trace: B2 frame 182

Target (2,2,-2) at UTC 1791674187.681857. Nearby PX4 estimate: (1.985,1.743,-1.999) m. OFFBOARD and armed.

Native body: (2.035,1.695,-10.042). Camera: (2.335,1.684,-10.193). Their map origin differs from PX4 local NED.

Frame clock 239.352. Camera clock 239.392. Body clock 239.405. Receiver-near illustration only. Acquisition pose is unknown.



Camera results and authored defects

The degree field received radians. Intended yaw 15° and pitch 10° became about 0.262° and 0.172°.

Our local angle and duration fixes have not run on Windows. The unexported duration-only retest still used wrong angles.

Repeatability and projected-feature criteria remain incomplete. Native pose metadata does not establish calibrated acquisition pose.



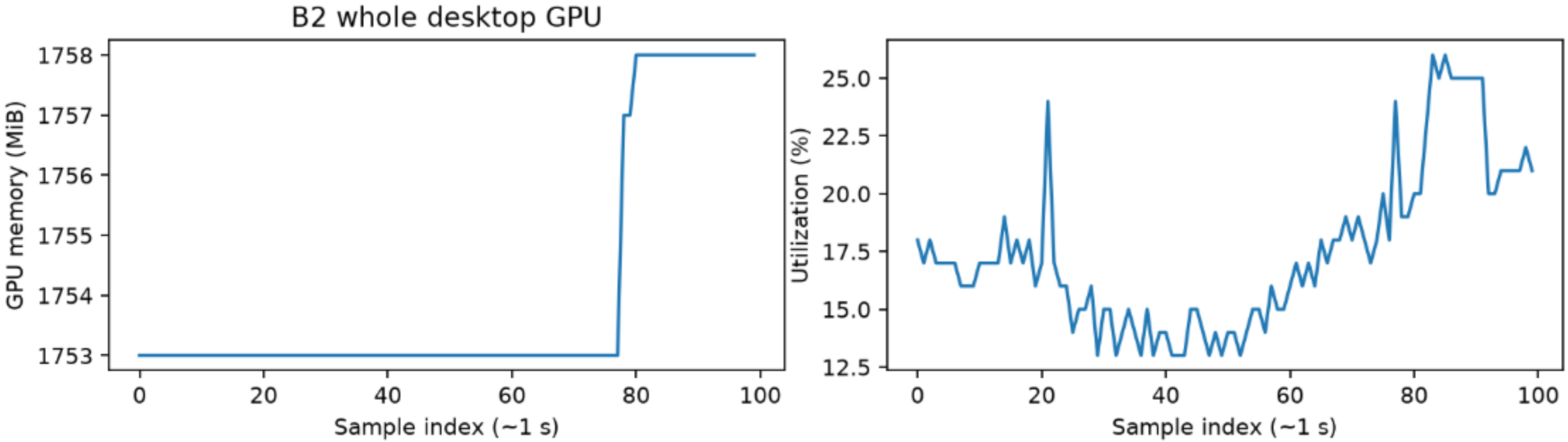
Native calibration and clock limits

Initial camera minus body was (0.3,0,-0.15) m despite the requested (0.3,0,0) mount. Actor-origin explanation is unresolved.

FOV axis, projection, optical axes, color/orientation and achieved extrinsics need independent checks for native calibrated images.

Acquisition UTC is unknown. A2 began with an old-looking image clock. No latency or geometric pass follows from receiver timestamps.

Hardware and measurement coverage



L4 / Windows Server 2025 / WSL1 worked on this host. 8 logical CPUs, 32 GB RAM, 400 GB SSD.

Whole-window GPU peaks: B1 1,761 MiB, B2 1,758 MiB. Mean utilization: 16.93% and 17.47%.

Desktop and Moonlight share the samples. B2 coverage ends mid-flight. Minimum VRAM, total RAM and renderer-only throughput are unknown.

Cleanup lifecycle and evidence boundaries

Flight records confirm landing and disarming in both B runs. Separate cleanup evaluators also returned confirmation summaries.

Our cleanup workers stayed alive until the 15 s deadline, exit -15. Raw cleanup logs are empty. The queue/finalizer cause remains unresolved.

Local lifecycle fixes need a Windows retest only if the unattended helper will be reused. This issue does not invalidate the retained flight passes.

R4R reuse options

Native visual baseline	**Demonstrated**	Stock RGB and control integration. Native calibration needed for geometric comparison.
Motion to external Gaussian renderer	**Proposed small adapter**	Verify chosen body frame/origin/scale. External renderer can supply its own camera and time.
Reconstructed mesh through UE	**Documented, eligibility unresolved**	Cooked UE assets plus XML. Verify installed loader and Free custom-scene permission.
Future visual controller loop	**Substantial integration**	Deliver external images to a controller, then commands to PX4. Scheduling/reset and task interfaces remain.

Applicability and research inputs

Residential, suburban and campus views below 100 m remain fidelity research questions. No real reconstruction entered this audit.

Sparse capture means missing view coverage. Sparse scene content means limited physical/perceptual structure. They need different evaluations.

Indoor navigation and racing are untested tasks. C lacks a compatible released policy path. That does not erase component usefulness.

Preservation and reviewer navigation

Mac copies verify all three evidence ZIPs and their internal manifests. Main records include 1,237 native PNGs and B1 ULog.

Standalone videos, the revised deck/PDF and an explanatory index publish to the existing private Drive folder. Sharing stays unchanged.

Final transcript, B2 ULog, strict-duration A1 retest and final scene/model/launcher inventory remain missing from durable delivery.

Minimal follow-up and interface learning

No restart is needed for the control/native-baseline decision. Optional recovery of missing files would take about 15–20 minutes.

Only measure the selected role next: external motion mapping, native calibrated replay, or our cleanup helper. Do not repeat every passed flight.

Understand NED/FRD versus camera axes, estimate versus truth, ACK versus achieved state, distinct clocks, and SITL versus physical HITL.